Word Count

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hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~$ mkdir WordCount
cd WordCount
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCMapper.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCReducer.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ nano WCDriver.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ export HADOOP_CLASSPATH=$(hadoop classpath)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ javac -classpath $HADOOP_CLASSPATH -d . WCMapper.java WCReducer.java WCDriver.java
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ jar -cvf wordcount.jar *.class
added manifest
adding: WCDriver.class(in = 1397) (out= 689)(deflated 50%)
adding: WCMapper.class(in = 1857) (out= 757)(deflated 59%)
adding: WCReducer.class(in = 1529) (out= 605)(deflated 60%)
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ cd /usr/local/hadoop/sbin
start-all.sh
jps
bash: cd: /usr/local/hadoop/sbin: No such file or directory
WARNING: Attempting to start all Apache Hadoop daemons as hadoop in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [bmscecse-HP-Elite-Tower-600-G9-Desktop-PC]
Starting resourcemanager
Starting nodemanagers
9152 Jps
8274 SecondaryNameNode
7977 DataNode
7802 NameNode
8570 ResourceManager
8732 NodeManager
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ mkdir input
nano input/sample.txt
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
2026-04-24 14:17:12,433 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-24 14:17:12,470 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-24 14:17:12,477 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,533 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-04-24 14:17:12,579 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-24 14:17:12,607 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local1387788888_0001
2026-04-24 14:17:12,666 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-24 14:17:12,720 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-24 14:17:12,720 INFO mapreduce.Job: Running job: job_local1387788888_0001
2026-04-24 14:17:12,721 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-24 14:17:12,722 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks
```

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hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -mkdir /inputbda
hadoop fs -put input/sample.txt /inputbda
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop jar wordcount.jar WCDriver /inputbda/sample.txt /outputbda
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2026-04-24 14:17:12,725 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,762 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-24 14:17:12,764 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,775 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,780 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-04-24 14:17:12,784 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,793 INFO mapred.MapTask: numReduceTasks: 1
2026-04-24 14:17:12,821 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-04-24 14:17:12,821 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100
2026-04-24 14:17:12,821 INFO mapred.MapTask: soft limit at 83886080
2026-04-24 14:17:12,821 INFO mapred.MapTask: bufstart = 0; bufvoid = 104857600
2026-04-24 14:17:12,821 INFO mapred.MapTask: kvstart = 26214396; length = 6553600
2026-04-24 14:17:12,823 INFO mapred.MapTask: Map output collector class = org.apache.hadoop.mapred.MapTask$MapOutputBuffer
2026-04-24 14:17:12,884 INFO mapred.LocalJobRunner:
2026-04-24 14:17:12,884 INFO mapred.MapTask: Starting flush of map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: Spilling map output
2026-04-24 14:17:12,884 INFO mapred.MapTask: bufstart = 0; bufend = 169; bufvoid = 104857600
2026-04-24 14:17:12,884 INFO mapred.MapTask: kvstart = 26214396(104857584); kvend = 26214320(104857280); length = 77/6553600
2026-04-24 14:17:12,887 INFO mapred.MapTask: Finished spill 0
2026-04-24 14:17:12,892 INFO mapred.Task: Task:attempt_local1387788888_0001_m_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,894 INFO mapred.LocalJobRunner: hdfs://localhost:9000/inputbda/sample.txt:0+89
2026-04-24 14:17:12,894 INFO mapred.Task: Task 'attempt_local1387788888_0001_m_000000_0' done.
2026-04-24 14:17:12,897 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_m_000000_0: Counters: 23
        File System Counters
                FILE: Number of bytes read=2920
                FILE: Number of bytes written=644387
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=0
                HDFS: Number of read operations=5
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=1
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=5
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                Map input records=5
                Map output records=20
                Map output bytes=169
                Map output materialized bytes=215
                Input split bytes=93
                Combine input records=0
                Spilled Records=20
                Failed Shuffles=0
                Merged Map outputs=0
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=526385152
        File Input Format Counters
                Bytes Read=89
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,897 INFO mapred.LocalJobRunner: map task executor complete.
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Waiting for reduce tasks
2026-04-24 14:17:12,899 INFO mapred.LocalJobRunner: Starting task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-24 14:17:12,902 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-24 14:17:12,902 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-04-24 14:17:12,904 INFO mapred.ReduceTask: Using ShuffleConsumerPlugin: org.apache.hadoop.mapreduce.task.reduce.Shuffle@53ac8b1c
2026-04-24 14:17:12,904 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-24 14:17:12,911 INFO reduce.MergeManagerImpl: MergerManager: memoryLimit=5830921216, maxSingleShuffleLimit=1457730304, mergeThreshold=3848408064, ioSortFactor=10, memToMemMergeOutputsThreshold=
2026-04-24 14:17:12,912 INFO reduce.EventFetcher: attempt_local1387788888_0001_r_000000_0 Thread started: EventFetcher for fetching Map Completion Events
2026-04-24 14:17:12,925 INFO reduce.LocalFetcher: localfetcher#1 about to shuffle output of map attempt_local1387788888_0001_m_000000_0 decomp: 211 len: 215 to MEMORY
2026-04-24 14:17:12,926 INFO reduce.InMemoryMapOutput: Read 211 bytes from map-output for attempt_local1387788888_0001_m_000000_0
2026-04-24 14:17:12,927 INFO reduce.MergeManagerImpl: closeInMemoryFile -> map-output of size: 211, inMemoryMapOutputs.size() -> 1, commitMemory -> 0, usedMemory ->211
2026-04-24 14:17:12,927 INFO reduce.EventFetcher: EventFetcher is interrupted.. Returning
2026-04-24 14:17:12,928 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,928 INFO reduce.MergeManagerImpl: finalMerge called with 1 in-memory map-outputs and 0 on-disk map-outputs
2026-04-24 14:17:12,930 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,930 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merged 1 segments, 211 bytes to disk to satisfy reduce memory limit
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 1 files, 215 bytes from disk
2026-04-24 14:17:12,931 INFO reduce.MergeManagerImpl: Merging 0 segments, 0 bytes from memory into reduce
2026-04-24 14:17:12,931 INFO mapred.Merger: Merging 1 sorted segments
2026-04-24 14:17:12,931 INFO mapred.Merger: Down to the last merge-pass, with 1 segments left of total size: 205 bytes
2026-04-24 14:17:12,931 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,991 INFO mapred.Task: Task:attempt_local1387788888_0001_r_000000_0 is done. And is in the process of committing
2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=3382
                FILE: Number of bytes written=644602
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
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2026-04-24 14:17:12,992 INFO mapred.LocalJobRunner: 1 / 1 copied.
2026-04-24 14:17:12,992 INFO mapred.Task: Task attempt_local1387788888_0001_r_000000_0 is allowed to commit now
2026-04-24 14:17:13,007 INFO output.FileOutputCommitter: Saved output of task 'attempt_local1387788888_0001_r_000000_0' to hdfs://localhost:9000/outputbda
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce > reduce
2026-04-24 14:17:13,008 INFO mapred.Task: Task 'attempt_local1387788888_0001_r_000000_0' done.
2026-04-24 14:17:13,008 INFO mapred.Task: Final Counters for attempt_local1387788888_0001_r_000000_0: Counters: 30
        File System Counters
                FILE: Number of bytes read=3382
                FILE: Number of bytes written=644602
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=89
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=10
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=3
                HDFS: Number of bytes read erasure-coded=0
        uce Framework
                Combine input records=0
                Combine output records=0
                Reduce input groups=10
                Reduce shuffle bytes=215
                Reduce input records=20
                Reduce output records=10
                Spilled Records=20
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=526385152
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Output Format Counters
                Bytes Written=69
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: Finishing task: attempt_local1387788888_0001_r_000000_0
2026-04-24 14:17:13,008 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-04-24 14:17:13,723 INFO mapreduce.Job: Job job_local1387788888_0001 running in uber mode : false
2026-04-24 14:17:13,723 INFO mapreduce.Job:  map 100% reduce 100%
2026-04-24 14:17:13,724 INFO mapreduce.Job: Job job_local1387788888_0001 completed successfully
2026-04-24 14:17:13,728 INFO mapreduce.Job: Counters: 36
        File System Counters
                FILE: Number of bytes read=6302
                FILE: Number of bytes written=1288989
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=178
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=15
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Ubuntu Software

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                FILE: Number of bytes written=1288989
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
                HDFS: Number of bytes read=178
                HDFS: Number of bytes written=69
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=5
                Map output records=20
                Map output bytes=169
                Map output materialized bytes=215
                Input split bytes=93
                Combine input records=0
                Combine output records=0
                Reduce input groups=10
                Reduce shuffle bytes=215
                Reduce input records=20
                Reduce output records=10
                Spilled Records=40
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=0
                Total committed heap usage (bytes)=1052770304
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=89
        File Output Format Counters
                Bytes Written=69
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -ls /outputbda
Found 2 items
-rw-r--r--   1 hadoop supergroup          0 2026-04-24 14:17 /outputbda/_SUCCESS
-rw-r--r--   1 hadoop supergroup         69 2026-04-24 14:17 /outputbda/part-00000
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$ hadoop fs -cat /outputbda/part-00000
are     1
brother 1
family  1
hi      1
how     5
is      4
job     1
sister  1
you     1
your    4
hadoop@bmscecse-HP-Elite-Tower-600-G9-Desktop-PC:~/WordCount$
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